Research Programme on the Mathematization of Karl Marx's *Capital*

Unified Treatise on Volume I, Parts I–III (Chapters 1–11):
Commodity Axiomatics, Stochastic Circulation, and the Valorization Operator

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Abstract

This treatise establishes a unified mathematical formalization of Karl Marx's *Capital*, Volume I, Chapters 1 through 11, organized into three interconnected modules. In **Module 1 (Chapter 1)**, we construct the dual space of the commodity via graded monoids and probability measure spaces, define the Real Abstraction Operator mapping concrete labor distributions to Socially Necessary Labor Time (SNLT), formalize the non-symmetric Relative-Equivalent Operator, model Commodity Fetishism as a quotient projection over a hidden fiber bundle of social labor relations, and prove the endogenous emergence of the General Equivalent via symmetry-breaking potential game dynamics on the hollow row-stochastic manifold. In **Module 2 (Chapters 2–3)**, we formalize the exchange process and simple circulation ($C-M-C'$), modeling the *salto mortale* (realization filter), deriving Marx's Law of the Quantity of Circulating Money, and proving a Saddle-Node Crisis Bifurcation Theorem induced by hoarding and liquidity panic. In **Module 3 (Chapters 4–11)**, we formalize the General Formula for Capital ($M-C-M'$), prove the Conservation of Value in Pure Circulation under arbitrary skew-symmetric bilateral markups, resolve the conservation contradiction via stock-flow operator splitting between Labor-Power and Living Labor, and formulate the differential geometry of the working day and Marx's Three Laws of the Mass of Surplus-Value.

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Part I

Module 1: Axiomatics of the Commodity-Form, Real Abstraction, and Graph Dynamics of the General Equivalent (Volume I, Chapter 1)

1 Introduction: Beyond Static Linearism

The mathematical formalization of Marxian political economy has historically been dominated by the neo-Ricardian or Sraffian tradition (e.g., Bortkiewicz, Seton, Morishima, Okishio), which models production through non-negative Leontief matrices and Perron-Frobenius spectral theory. While powerful for analyzing intersectoral reproduction and the transformation problem of Volume III, linear input-output algebra bypasses the foundational logical movement of Volume I, Chapter 1 (*"The Commodity"*).

In Chapter 1, Marx does not presuppose a pre-existing scalar metric of price or commensurability. Instead, he investigates how a **homogeneous metric space (Value)** and a **universal canonical sink (Money)** emerge endogenously out of a **heterogeneous space of incommensurable physical objects (Use-Values) and private concrete labors**. Formalizing this genesis requires mathematical structures capable of representing qualitative incommensurability, real abstraction, polar asymmetry, and topological symmetry-breaking.

2 The Dual Space of the Commodity (Chapter 1, Sections 1 & 2)

A commodity $c \in \mathcal{C}$ is a dual entity possessing **Use-Value** (qualitative, incommensurable) and **Value** (quantitative, commensurable). We represent this duality by constructing two distinct spaces linked by a social abstraction functor.

2.1 The Physical Use-Value Space ($\mathcal{U}$)

Let $\mathcal{T} = \{1, 2, \dots, n\}$ be the index set of distinct concrete commodity types (e.g., linen, coats, iron, corn). Because distinct use-values lack a common physical metric (1 coat + 1 ton of iron admits no physical scalar sum), the space of physical commodities is constructed as a **disjoint union of 1-dimensional positive cones** (a Graded Monoid):

$$\mathcal{U} = \bigsqcup_{i \in \mathcal{T}} \mathcal{U}_i, \quad \text{where } \mathcal{U}_i = \mathbb{R}_{\geq 0} \cdot \mathbf{e}_i \tag{1}$$

Here, $\mathbf{e}_i$ denotes the basis unit of physical measurement for commodity i (yards, tons, bushels). A specific commodity bundle of type i is written as $q_i \mathbf{e}_i \in \mathcal{U}_i$. There is no native inner product, norm, or addition operator on $\mathcal{U}$ that directly combines or compares $\mathbf{e}_i$ and $\mathbf{e}_j$ for $i \neq j$.

2.2 Concrete vs. Abstract Labor and Socially Necessary Labor Time (SNLT)

Let $\mathcal{L}_i$ represent the set of individual producers (or production processes) creating commodity i. Each producer $\omega \in \mathcal{L}_i$ expends a specific **concrete labor** characterized by technical vectors (skill, intensity, tool efficiency) and requires individual physical clock-time $\tau_i(\omega) \in \mathbb{R}_{>0}$ to produce one physical unit $\mathbf{e}_i$.

Let $(\mathcal{L}_i, \Sigma_i, \mu_i)$ be a probability measure space over the producers of sector i, where $\mu_i(\omega)$ represents the share of total social output of commodity i produced by agent ω, satisfying $\int_{\mathcal{L}_i} d\mu_i(\omega) = 1$.

Definition 2.1 (The Real Abstraction Operator $\mathcal{A}$)**.** *Commodity exchange strips away the qualitative specificity of concrete labor, projecting heterogeneous labor processes onto a single one-dimensional scalar manifold:* ***Abstract Human Labor*** *($\mathcal{V} \cong \mathbb{R}_{\geq 0}$). The unit value λ_i of commodity i—its* ***Socially Necessary Labor Time (SNLT)****—is defined via the expectation operator over the sectoral productivity distribution, weighted by skill/intensity multipliers $\sigma_i(\omega) \in \mathbb{R}_{>0}$:*

$$\lambda_i = \mathcal{A}(\mathcal{L}_i) \coloneqq \int_{\mathcal{L}_i} \sigma_i(\omega)\tau_i(\omega)\, d\mu_i(\omega) \tag{2}$$

Remark 2.1 (Comparative Statics of Technological Shock)**.** *If a technological revolution occurs (e.g., power looms replace handlooms), the measure μ_i shifts mass toward producers with lower $\tau_i(\omega)$. This immediately depresses λ_i across the entire existing inventory of commodity i, capturing Marx's insistence that value is a contemporaneous social relation rather than a historical accounting of past effort.*

The **Total Value Functional** $V : \mathcal{U} \to \mathbb{R}_{\geq 0}$ acting on a homogeneous commodity bundle $q_i\mathbf{e}_i \in \mathcal{U}_i$ is therefore:

$$V(q_i\mathbf{e}_i) = q_i\lambda_i \tag{3}$$

3 The Algebra of the Value-Form (Chapter 1, Section 3)

Marx demonstrates that value (λ_i) cannot appear directly as an isolated scalar; a commodity cannot express its value in its own physical body. Value requires a **polar relational mapping** between two distinct commodities.

3.1 The Value Equivalence Relation ($\sim_V$)

Define a binary relation $\sim_V$ on the disjoint space $\mathcal{U}$:

$$q_i\mathbf{e}_i \sim_V q_j\mathbf{e}_j \iff V(q_i\mathbf{e}_i) = V(q_j\mathbf{e}_j) \iff q_i\lambda_i = q_j\lambda_j \tag{4}$$

While $\sim_V$ is mathematically an equivalence relation (reflexive, symmetric, transitive) on the underlying scalar space $\mathcal{V}$, **its phenomenal expression in the commodity world is strictly non-symmetric**. In the value equation "x commodity A is worth y commodity B," A and B occupy polarized structural roles.

3.2 The Non-Symmetric Value Expression Operator ($\mathcal{E}$)

Definition 3.1 (Relative-Equivalent Operator)**.** *For $i \neq j$, define the operator $\mathcal{E} : \mathcal{U}_i \times \{\mathbf{e}_j\} \to \mathcal{U}_j$, where commodity i occupies the* ***Relative Form*** *(the active subject seeking value-expression) and commodity j occupies the* ***Equivalent Form*** *(the passive physical mirror whose bodily use-value measures i):*

$$\mathcal{E}(q_i\mathbf{e}_i \mid \mathbf{e}_j) = \left(q_i\frac{\lambda_i}{\lambda_j}\right)\mathbf{e}_j \in \mathcal{U}_j \tag{5}$$

Two fundamental formal properties directly encode Marx's dialectical exposition:

(i) **Irreflexivity of Phenomenal Expression:** For $i = j$, $\mathcal{E}(q_i\mathbf{e}_i \mid \mathbf{e}_i) = q_i\mathbf{e}_i$, which reduces to a meaningless tautology (20 yards of linen $=$ 20 yards of linen) and fails to express value. Valid value-expressions are strictly restricted to off-diagonal pairs $(i, j) \in \mathcal{T} \times \mathcal{T}$ with $i \neq j$.

(ii) **Dimensional Projection:** The codomain of $\mathcal{E}(q_i\mathbf{e}_i \mid \mathbf{e}_j)$ is the physical subspace $\mathcal{U}_j$, *not* abstract labor hours $\mathcal{V}$. Abstract value λ_i becomes phenomenally visible only by projecting i onto the concrete physical metric of j.

3.3 Topological Evolution of the Four Forms of Value

We formalize the progression in Chapter 1, Section 3 as a sequence of **directed graphs** $G_k = (\mathcal{T}, E_k)$, where a directed edge $(i \to j) \in E_k$ signifies that commodity i (Relative Form) expresses its value in the physical body of commodity j (Equivalent Form):

Form of Value	Graph Topology G_k	Algebraic Expression	Structural Defect
I. Elementary / Accidental	**Isolated Dyad:** $E_{\mathrm{I}} = \{(i \to j)\}$	$q_i\mathbf{e}_i \xrightarrow{\mathcal{E}} \left(q_i \frac{\lambda_i}{\lambda_j}\right)\mathbf{e}_j$	*Local & Contingent:* Pairwise ratio only; no global transitivity across $\mathcal{T}$.
II. Total / Expanded	**Out-Star Graph:** $E_{\mathrm{II}} = \{(i \to k) \mid k \neq i\}$	$\mathbf{E}_{\mathrm{II}}(q_i\mathbf{e}_i) = \bigoplus_{k\neq i}\left(q_i \frac{\lambda_i}{\lambda_k}\right)\mathbf{e}_k$	*Incomplete & Heterogeneous:* Non-convergent direct sum of orthogonal metrics.
III. General Form	**In-Star Graph (Sink):** $E_{\mathrm{III}} = \{(k \to u) \mid k \neq u\}$	$\forall k \neq u:$ $\mathcal{E}(q_k\mathbf{e}_k \mid \mathbf{e}_u) = \left(q_k \frac{\lambda_k}{\lambda_u}\right)\mathbf{e}_u$	*Symmetry Degeneracy:* Any $u \in \mathcal{T}$ can act as sink; competing sinks block unity.
IV. Money-Form	**Symmetry-Broken Sink:** Unique $m \in \mathcal{T}$ (Gold)	$p_k = \frac{\lambda_k}{\lambda_m}$ (Price scalar in $\mathbf{e}_m$)	*Ideal vs. Realized Value:* Bridges Ch. 1 to Ch. 2–3 (*salto mortale*).

Dialectical Dualization (Form II → Form III): In graph-theoretic and categorical terms, Form III is the **transpose (dual graph)** of Form II: $G_{\mathrm{III}} = G_{\mathrm{II}}^{\top}$. By inverting the rays of the Out-Star so that all $k \in \mathcal{T} \setminus \{u\}$ point into a single excluded commodity u, the disjoint space $\mathcal{U}$ acquires a canonical isomorphism to $\mathbb{R}_{\geq 0}$ mediated by the physical basis unit $\mathbf{e}_u$.

4 Commodity Fetishism as a Quotient Topology (Chapter 1, Section 4)

In Section 4, Marx introduces **Commodity Fetishism**: the social relations among private producers $(\mathcal{L}_1, \ldots, \mathcal{L}_n)$ do not appear directly as transparent social relations; instead, they appear exclusively as quantitative exchange-ratios (prices) attached to things.

We formalize this via an **epimorphism (projection) and a hidden fiber bundle**:

1. Let $\Omega = \prod_{i\in\mathcal{T}} \mathcal{L}_i$ be the **Total Social Labor Configuration Space**, encoding the complete network of the social division of labor, concrete production relations, and productivity distributions μ_i.

2. Let $\mathcal{P} = \mathbb{RP}^{n-1}_{>0}$ be the **Projective Price Simplex** of observable exchange ratios $[p_1 : p_2 : \cdots : p_n]$, where $p_i = \lambda_i/\lambda_m$.

3. The market exchange mechanism acts as a projection map $\Pi : \Omega \to \mathcal{P}$.

Because $\dim(\Omega) \gg \dim(\mathcal{P})$, Π is a non-injective surjection with high-dimensional fibers $\Pi^{-1}(\mathbf{p})$. Private producers in a commodity-producing society cannot observe Ω directly; they receive feedback signals exclusively from $\mathcal{P}$ (fluctuations in relative prices) and reallocate their concrete labor accordingly. **Fetishism** is the structural illusion induced by the projection Π, whereby invariants of the fiber space Ω (social relations of production) are misattributed to the base manifold $\mathcal{P}$ (natural properties of physical commodities).

5 Directed Graph Dynamics: Proving the Emergence of the General Equivalent

In Chapter 1, Section 3 (and Chapter 2, "The Process of Exchange"), Marx identifies a fundamental logical and practical contradiction between the **Expanded Form of Value ($G_{\mathbf{II}}$)** and the requirements of a coherent commodity-producing society. If every commodity simultaneously attempts to exclude all others as its private out-star, no commodity is excluded, and social commensurability collapses into $O(n^2)$ non-transitive bilateral ratios. Consequently, the "social action of all other commodities" (*gesellschaftliche Tat*) must spontaneously break the symmetry of the commodity space $\mathcal{T}$ and isolate a unique sink $u^* \in \mathcal{T}$—transforming the collection of Out-Stars (G_{II}) into a single **In-Star topology ($G_{\mathbf{III}}$)**.

5.1 The Hollow Row-Stochastic Manifold of Value-Expression Networks

Let $\mathcal{T} = \{1, 2, \ldots, n\}$ ($n \geq 3$) be the set of distinct commodity types, with positive unit values $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_n) \in \mathbb{R}^n_{>0}$.

Definition 5.1 (Value-Expression Adjacency Manifold)**.** *At time $t \geq 0$, each commodity sector $i \in \mathcal{T}$ allocates weights $w_{ij}(t) \in [0, 1]$ across the other $n-1$ commodities to serve as its **Equivalent Form**. Because the Relative-Equivalent Operator $\mathcal{E}(q_i\mathbf{e}_i \mid \mathbf{e}_j)$ is strictly irreflexive ($i \neq j$), the weighted directed graph $G(t) = (\mathcal{T}, W(t))$ evolves on the **hollow row-stochastic manifold**:*

$$\mathcal{W} = \left\{ W \in \mathbb{R}^{n \times n}_{\geq 0} \;\middle|\; w_{ii} = 0 \text{ for all } i \in \mathcal{T}, \text{ and } \sum_{j \neq i} w_{ij} = 1 \text{ for all } i \in \mathcal{T} \right\} \tag{6}$$

Definition 5.2 (Equivalent Prominence / Normalized In-Degree)**.** *For any state $W \in \mathcal{W}$, define the **Equivalent Prominence** of commodity $j \in \mathcal{T}$ as:*

$$d_j(W) \coloneqq \frac{1}{n-1} \sum_{k \neq j} w_{kj} \in [0, 1] \tag{7}$$

5.2 Micro-Foundations and the Global Social Commensurability Potential

When commodity i expresses its value in commodity j ($i \to j$), that expression achieves social commensurability to the extent that other commodities $k \notin \{i, j\}$ also recognize j as their equivalent ($k \to j$). When both $i \to j$ and $k \to j$ hold, commodity j mediates a convergent two-step wedge $i \to j \leftarrow k$ in the same physical subspace $\mathcal{U}_j$:

$$\frac{\mathcal{E}(q_i\mathbf{e}_i \mid \mathbf{e}_j)}{\mathcal{E}(q_k\mathbf{e}_k \mid \mathbf{e}_j)} = \frac{q_i\lambda_i/\lambda_j}{q_k\lambda_k/\lambda_j} = \frac{q_i\lambda_i}{q_k\lambda_k} \tag{8}$$

Thus, the strategic payoff to sector i from allocating weight w_{ij} to equivalent j is given by:

$$u_{ij}(W) = \alpha \sum_{k \notin \{i,j\}} w_{kj} + \eta_j \tag{9}$$

where $\alpha > 0$ measures the **social commensurability coupling** and $\eta_j \geq 0$ represents physical suitability for the equivalent role (divisibility, durability, uniformity).

Lemma 5.1 (Existence of the Social Commensurability Potential)**.** *The n-sector value-expression system on $\mathcal{W} = \prod_{i=1}^{n} \Delta_i$ is an **Exact Potential Game** with global potential functional $\Phi : \mathcal{W} \to \mathbb{R}$ defined by:*

$$\Phi(W) \coloneqq \frac{\alpha}{2} \sum_{j=1}^{n} \sum_{i \neq j} \sum_{k \notin \{i,j\}} w_{ij} w_{kj} + \sum_{j=1}^{n} \eta_j \sum_{i \neq j} w_{ij} = \frac{\alpha}{2} \sum_{j=1}^{n} \left[\left(\sum_{i \neq j} w_{ij} \right)^2 - \sum_{i \neq j} w_{ij}^2 \right] + \sum_{j=1}^{n} \eta_j \left(\sum_{i \neq j} w_{ij} \right) \tag{10}$$

satisfying $\frac{\partial \Phi(W)}{\partial w_{ij}} = u_{ij}(W)$ for all $i \neq j$.

Proof. Direct partial differentiation of $\Phi(W)$ with respect to w_{ij} $(i \neq j)$ isolates the j-th summand and yields $\frac{\partial \Phi(W)}{\partial w_{ij}} = \alpha \sum_{k \notin \{i,j\}} w_{kj} + \eta_j = u_{ij}(W)$. □

5.3 Replicator Dynamics and the Symmetry-Breaking Theorem

Each commodity sector $i \in \mathcal{T}$ updates its equivalent weights $w_{ij}(t)$ via continuous-time multi-population replicator dynamics:

$$\dot{w}_{ij} = w_{ij} \left(u_{ij}(W) - \bar{u}_i(W) \right), \quad \forall i \in \mathcal{T},\ j \neq i, \quad \text{where } \bar{u}_i(W) = \sum_{l \neq i} w_{il} u_{il}(W) \tag{11}$$

Theorem 5.1 (Instability of the Expanded Form G_{II} and Global Convergence to the General Form G_{III})**.** *Assume pure initial symmetry among commodities ($\eta_1 = \eta_2 = \cdots = \eta_n = 0$) and $\alpha > 0$. Under the replicator dynamics on $\mathcal{W}$:*

(1) ***Strict Lyapunov Ascent:*** *$\Phi(W(t))$ is strictly non-decreasing along trajectories:*

$$\frac{d}{dt} \Phi(W(t)) = \sum_{i=1}^{n} \sum_{j \neq i} w_{ij} \left(u_{ij}(W) - \bar{u}_i(W) \right)^2 \geq 0 \tag{12}$$

with equality if and only if W is a stationary point.

(2) ***Linear Instability of the Expanded Form (W^{II}):*** *The symmetric Expanded Form centroid W^{II} ($w_{ij} = \frac{1}{n-1}$) is dynamically unstable. Any perturbation $S_j = \sum_{i \neq j} \epsilon_{ij}$ to the total in-degree of commodity j grows exponentially with strictly positive eigenvalue:*

$$\mu_{star} = \alpha \left(\frac{n^2 - 3n + 1}{(n-1)^2} \right) > 0 \quad (\forall n \geq 3) \tag{13}$$

(3) ***Global Topological Collapse to the In-Star (G_{III}):*** *For almost all initial conditions $W(0) \in \mathrm{int}(\mathcal{W})$, the trajectory $W(t)$ converges as $t \to \infty$ to a corner configuration $W^{III(u^*)}$ where a single commodity $u^* \in \mathcal{T}$ emerges as the unique General Equivalent:*

$$\lim_{t \to \infty} w_{iu^*}(t) = 1 \quad \forall i \in \mathcal{T} \setminus \{u^*\}, \quad \textit{and} \quad \lim_{t \to \infty} d_{u^*}(W(t)) = 1 \tag{14}$$

Proof. **Part (1):** By the chain rule and Lemma 5.1, $\dot{\Phi} = \sum_{i=1}^{n} \sum_{j \neq i} u_{ij} w_{ij} (u_{ij} - \bar{u}_i)$. Subtracting $\sum_{i=1}^{n} \bar{u}_i \sum_{j \neq i} w_{ij}(u_{ij} - \bar{u}_i) = 0$ yields $\dot{\Phi} = \sum_{i=1}^{n} \sum_{j \neq i} w_{ij} (u_{ij} - \bar{u}_i)^2 \geq 0$. By compactness of $\mathcal{W}$ and LaSalle's Invariance Principle, all trajectories converge to the stationary locus $\{\dot{\Phi} = 0\}$.

Part (2): Linearizing around $w_{ij}^{\mathrm{II}} = \frac{1}{n-1}$ with $w_{ij} = \frac{1}{n-1} + \epsilon_{ij}$ ($\epsilon_{ii} = 0$, $\sum_{j\neq i} \epsilon_{ij} = 0$) and defining $S_j = \sum_{k\neq j} \epsilon_{kj}$ ($\sum_{j=1}^{n} S_j = 0$), the perturbed payoff is $\delta u_{ij} = \alpha(S_j - \epsilon_{ij})$. Summing the linearized replicator equation over $i \neq j$ gives:

$$\dot{S}_j = \frac{\alpha}{n-1} \sum_{i\neq j} \left[(S_j - \epsilon_{ij}) - \frac{1}{n-1} \sum_{l\neq i} (S_l - \epsilon_{il}) \right] = \frac{\alpha}{n-1} \left[(n-2)S_j - \frac{1}{n-1} S_j \right] = \alpha \left(\frac{n^2 - 3n + 1}{(n-1)^2} \right) S_j \tag{15}$$

Since $n^2 - 3n + 1 \geq 1 > 0$ for all $n \geq 3$, $\mu_{\mathrm{star}} > 0$, proving instability of W^{II}.

Part (3): A rest point W^* is asymptotically stable under replicator dynamics of a potential game if and only if it is a strict local maximizer of $\Phi(W)$. Along any coordinated multi-agent interior perturbation where two agents i_1, i_2 shift weight $\pm\delta$ between two distinct equivalents j_1, j_2, the second variation of Φ is $\delta^2\Phi = 4\alpha\delta^2 > 0$. Thus, no interior or partially mixed multi-sink state can be a local maximum. At pure vertices where each i selects a unique target $f(i) \neq i$ with in-degrees $k_j = |f^{-1}(j)| \in \{0, \ldots, n-1\}$ ($\sum_j k_j = n$), $\Phi(W) = \frac{\alpha}{2} \sum_{j=1}^{n} k_j(k_j - 1)$. By strict convexity of $k(k-1)$, Φ is uniquely maximized when a single sink u^* achieves $k_{u^*} = n - 1$, one node $v = f(u^*)$ has $k_v = 1$, and all other $n-2$ nodes have $k_j = 0$, yielding $\max \Phi = \frac{\alpha}{2}(n-1)(n-2)$ at $W^{\mathrm{III}(u^*)}$. □

Corollary 5.1 (Polar Exclusion of the General Equivalent and Transition to the Money-Form G_{IV})**.** *At the General Form attractor $W^{III(u^*)}$:*

(i) ***Exclusion from the Relative Form:*** *For the General Equivalent u^* itself, $u_{u^*v}(W^{III(u^*)}) = \alpha \sum_{k\notin\{u^*,v\}} w_{kv} = 0$ for all $v \neq u^*$, so $\nabla_{\mathbf{w}_{u^*}} \Phi = \mathbf{0}$ (when $\boldsymbol{\eta} = \mathbf{0}$). Thus, u^* is structurally excluded from the active Relative Form; its value can only be read by transposing the In-Star G_{III} back into the Out-Star G_{II}.*

(ii) ***Symmetry Breaking to the Money-Form (G_{IV}):*** *When physical suitability $\boldsymbol{\eta}$ has a unique maximum $\eta_m > \eta_j$ ($\forall j \neq m$, e.g., Gold), the S_n permutation degeneracy is broken, uniquely selecting $u^* = m$ as the global maximizer $\Phi(W^{III(m)}) = \frac{\alpha}{2}(n-1)(n-2) + (n-1)\eta_m$ and establishing the canonical price vector $p_i = \lambda_i/\lambda_m$.*

Part II

Module 2: Stochastic Exchange, the *Salto Mortale*, and the Metamorphosis $C - M - C'$ (Volume I, Chapters 2 & 3)

6 The Mutual Exclusion Principle and Bipartite Exchange Polarity (Chapter 2)

In Chapter 1, unit value λ_i and money-price $p_i = \lambda_i/\lambda_m$ are derived as ideal expectations over the production measure μ_i. In Chapters 2 and 3, a commodity bundle $q_i\mathbf{e}_i$ enters the market bearing **potential (ideal) value** $V_i^{\text{pot}} = q_i\lambda_i$, which must be validated as actual social labor through the stochastic realization process $C_i \to M$ (*salto mortale*).

Definition 6.1 (The Agent-Commodity State Tuple)**.** *Let $\mathcal{A} = \{1, 2, \ldots, N\}$ be the population of private commodity owners. At time t, agent $\alpha \in \mathcal{A}$ is characterized by a state tuple:*

$$\mathcal{S}_\alpha(t) = (\mathbf{q}_\alpha(t),\, m_\alpha(t),\, \mathbf{u}_\alpha) \in \mathcal{U} \times \mathcal{U}_m \times \mathcal{B}(\mathcal{T}_0) \tag{16}$$

where $\mathbf{q}_\alpha(t) = q_{\alpha,i}\mathbf{e}_i \in \mathcal{U}_i$ is the physical commodity bundle produced by α, $m_\alpha(t) \in \mathbb{R}_{\geq 0} \cdot \mathbf{e}_m$ is agent α's holding of Money, and $\mathbf{u}_\alpha : \mathcal{T}_0 \to \{0, 1\}$ is a binary concrete need indicator over ordinary commodities $\mathcal{T}_0 = \mathcal{T} \setminus \{m\}$.

Axiom 6.1 (The Non-Use-Value Condition of Commodity Production)**.** *For any commodity producer $\alpha \in \mathcal{A}$ producing sector $i \in \mathcal{T}_0$:*

$$u_\alpha(i) = 0 \quad \textit{and} \quad \exists j \in \mathcal{T}_0 \setminus \{i\} \textit{ such that } u_\alpha(j) = 1 \tag{17}$$

That is, one's own product is a pure bearer of exchange-value (a non-use-value to its owner) and must be alienated to an agent β for whom $u_\beta(i) = 1$.

7 The *Salto Mortale* and the Realization Operator (Chapter 3, Section 2a)

In Chapter 3, Section 2a, Marx introduces the **second dimension of Socially Necessary Labor Time**: even if every unit of commodity i is produced in the technically necessary time λ_i, if aggregate effective social demand cannot absorb the total sector output $Q_i = \sum_{\alpha \in \mathcal{A}_i} q_{\alpha,i}$, the excess labor time expended in sector i fails to realize as social value.

Definition 7.1 (Produced Ideal Value vs. Effective Social Demand)**.** *For each sector $i \in \mathcal{T}_0$ over a market clearing period $[t, t + \Delta t]$:*

(i) ***Total Produced Ideal Value:*** *$V_i^{prod} = \lambda_i Q_i = \lambda_i \sum_{\alpha \in \mathcal{A}_i} q_{\alpha,i}$.*

(ii) ***Stochastic Effective Monetary Demand:*** *Let $M_i^{circ}(\omega_t) \in \mathbb{R}_{\geq 0}$ be a random variable on a filtered market probability space $(\Xi, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ representing the total quantity of money $\mathbf{e}_m$ directed by buyers toward sector i. In units of abstract labor time, the socially allocated demand is $D_i(\omega_t) = \lambda_m M_i^{circ}(\omega_t)$.*

Definition 7.2 (The *Salto Mortale* Realization Operator $\mathcal{R}_i$)**.** *The transition from ideal commodity value C_i to realized money M is governed by a non-linear operator $\mathcal{R}_i : \mathcal{U}_i \times \Xi \to \mathcal{U}_m$:*

(1) ***Quantity-Rationing Regime (Fixed Price*** $p_i = \lambda_i/\lambda_m$***):*** *Each unit of commodity i faces a realization survival probability:*

$$\pi_i(\omega_t) = \min\left(1,\ \frac{D_i(\omega_t)}{V_i^{prod}}\right) = \min\left(1,\ \frac{\lambda_m M_i^{circ}(\omega_t)}{\lambda_i Q_i}\right) \in [0,1] \tag{18}$$

yielding realized value $V_{\alpha,i}^{real}(\omega_t) = \lambda_m \mathcal{R}_i(q_{\alpha,i}\mathbf{e}_i, \omega_t) = \pi_i(\omega_t) q_{\alpha,i}\lambda_i \le V_{\alpha,i}^{pot}$*, with residual* $(1-\pi_i(\omega_t))q_{\alpha,i}\lambda_i$ *representing devalued labor time.*

(2) ***Market-Price Divergence Regime (Flexible Price*** $\tilde{p}_i(\omega_t) = M_i^{\mathbf{circ}}(\omega_t)/Q_i$***):*** *The market-validated unit labor value shifts to:*

$$\tilde{\lambda}_i(\omega_t) = \lambda_m \tilde{p}_i(\omega_t) = \lambda_i \left(\frac{D_i(\omega_t)}{V_i^{prod}}\right) \tag{19}$$

8 Operator Algebra of the Circuit $C_i - M - C_j$

Simple commodity circulation decomposes into two antithetical operator phases:

- **First Metamorphosis (Sale,** $C_i \to M$**):** The stochastic Alienation Operator $\hat{\mathcal{S}}_i : \mathcal{U}_i \to \mathcal{U}_m$:

$$\hat{\mathcal{S}}_i(q_i\mathbf{e}_i) = \left(\pi_i(\omega_t) q_i \frac{\lambda_i}{\lambda_m}\right)\mathbf{e}_m \tag{20}$$

- **Second Metamorphosis (Purchase,** $M \to C_j$**):** The Appropriation Operator $\hat{\mathcal{B}}_j : \mathcal{U}_m \to \mathcal{U}_j$:

$$\hat{\mathcal{B}}_j(m\mathbf{e}_m) = \left(m\frac{\lambda_m}{\lambda_j}\right)\mathbf{e}_j \tag{21}$$

Proposition 8.1 (Topological Asymmetry of Sale $\hat{\mathcal{S}}_i$ and Purchase $\hat{\mathcal{B}}_j$)**.** *In the Money-Form topology G_{IV} (In-Star with sink m), Sale ($\hat{\mathcal{S}}_i$) is a stochastic contraction ($\pi_i(\omega_t) \le 1$) because ordinary commodities $i \in \mathcal{T}_0$ occupy the Relative Form and lack direct universal exchangeability. Conversely, Purchase ($\hat{\mathcal{B}}_j$) traverses the transposed Out-Star $G_{IV}^{\top}$ from the Universal Equivalent sink m, possessing unconditional realizability ($\pi_m \equiv 1$ whenever inventories $Q_j > 0$ exist). The composite circuit operator $\hat{\mathcal{M}}_{i\to j} \coloneqq \hat{\mathcal{B}}_j \circ \hat{\mathcal{S}}_i : \mathcal{U}_i \to \mathcal{U}_j$ satisfies:*

$$\hat{\mathcal{M}}_{i\to j}(q_i\mathbf{e}_i) = \left(\pi_i(\omega_t) q_i \frac{\lambda_i}{\lambda_j}\right)\mathbf{e}_j, \quad \mathbb{E}\Big[V\Big(\hat{\mathcal{M}}_{i\to j}(q_i\mathbf{e}_i)\Big)\Big] \le V(q_i\mathbf{e}_i) \tag{22}$$

9 Breaking Say's Law: Markov Circulation Dynamics and the Crisis Bifurcation (Chapter 3, Sections 2b & 3)

Because $C_i \to M$ and $M \to C_j$ are separated in time by an intermediate holding state M, an agent who has sold ($C_i \to M$) is under no compulsion to immediately buy ($M \to C_j$). Money can immobilize itself as a **Hoard** (*Schatzbildung*).

Definition 9.1 (Four-State Circulation Markov Process)**.** *Social value in circulation transitions across four states $\Sigma = \{\mathbf{C}, \mathbf{M}_{circ}, \mathbf{M}_{hoard}, \mathbf{K}\}$ with macro-distribution $\mathbf{x}(t) = (x_C(t), x_M(t), x_H(t), x_K(t))^{\top}$:*

(1) **C** *(**Commodity Capital on Market**): Realizes into $\mathbf{M}_{circ}$ at rate $\gamma_{sale}(x_M) = \beta x_M$ ($\beta > 0$), or decays into crisis/default state **K** at expiration hazard rate $\delta_{crisis} > 0$.*

(2) $\mathbf{M}_{circ}$ *(**Active Circulating Medium**): Executes purchases $\mathbf{M}_{circ} \to \mathbf{C}$ at rate $\mu_{buy} > 0$, or withdraws into hoards $\mathbf{M}_{hoard}$ at rate $\sigma_{hoard} \ge 0$.*

(3) $\mathbf{M}_{hoard}$ ***(Hoarded Money / Liquidity Reserve)****: Re-enters active circulation* $\mathbf{M}_{hoard} \to \mathbf{M}_{circ}$ *at dishoarding rate* $\rho_{dishoard} > 0$.

(4) $\mathbf{K}$ ***(Crisis / Unrealized Default State)****: Absorbing or distressed inventory state resulting from failed realization.*

Proposition 9.1 (Marx's Law of the Quantity of Circulating Money)**.** *Let* $P_{sum}(t) = \sum_{i\in\mathcal{T}_0} p_i Q_i(t) = \frac{1}{\lambda_m}\sum_{i\in\mathcal{T}_0} \lambda_i Q_i(t)$ *be the sum of prices of commodities in circulation over period* T, $K_{credit}(t)$ *be mutual credit offsets,* $D_{due}(t)$ *be maturing debt obligations, and* v *be the velocity of money. Then the required active circulating money stock is:*

$$M^{active}(t) = \frac{P_{sum}(t) - K_{credit}(t) + D_{due}(t)}{v} = \frac{1}{v\lambda_m}\left(\sum_{i\in\mathcal{T}_0} \lambda_i Q_i(t) - \lambda_m K_{credit}(t) + \lambda_m D_{due}(t)\right) \tag{23}$$

Unlike neoclassical monetarism ($MV = PY$*), prices* $p_i = \lambda_i/\lambda_m$ *are determined prior to circulation by the production measure* μ_i*, and fluctuations in required* $M^{active}(t)$ *are buffered by the hoard reservoir* $M^{hoard}(t) = M^{total} - M^{active}(t)$.

Theorem 9.1 (Breakability of $C - M - C'$ and the Realization Crisis Phase Transition)**.** *Let total money be normalized to* $M^{total} = x_M(t) + x_H(t) = M_0$*. Under the hoarding dynamics* $\dot{x}_M = \rho_{dishoard}(M_0 - x_M) - \sigma_{hoard} x_M$*, the steady-state active money stock is* $x_M^*(h) = \frac{M_0}{1+h}$*, where* $h \coloneqq \sigma_{hoard}/\rho_{dishoard}$ *is the* ***Hoarding Propensity Ratio****, and the* ***Global Realization Efficiency*** *is:*

$$\mathcal{E}_{real}(h) = \frac{\gamma_{sale}(x_M^*)}{\gamma_{sale}(x_M^*) + \delta_{crisis}} = \frac{\beta M_0}{\beta M_0 + \delta_{crisis}(1+h)} \tag{24}$$

Suppose hoarding is endogenously coupled to realization failure via credit-default panic: $h(u) = h_0 + \tilde{\xi}u^2$*, where* $u \coloneqq 1 - \mathcal{E}_{real} \in [0,1]$ *is the unrealized crisis share and* $\tilde{\xi} = \xi/\rho_{dishoard} > 0$ *is panic sensitivity. Then:*

(i) Every equilibrium $u^* \in (0,1)$ *satisfies the fixed-point balance equation:*

$$\underbrace{\frac{\beta M_0}{\delta_{crisis}}\frac{u}{1-u}}_{L(u)} = \underbrace{1 + h_0 + \tilde{\xi}u^2}_{R(u)} \tag{25}$$

(ii) As $\tilde{\xi}$ *crosses the critical threshold* $\tilde{\xi}_c$ *defined by the tangency* $L(u_c) = R(u_c)$ *and* $L'(u_c) = R'(u_c)$*, the circulation system undergoes a* ***saddle-node bifurcation****, generating a stable* ***Acute Monetary Crisis / Liquidity Trap Attractor*** *(*$u_3^* \to 1$*,* $x_M^* \to 0$*) even prior to capitalist production.*

Proof. Substituting $x_M^* = \frac{M_0}{1+h(u)}$ into $u = 1 - \mathcal{E}_{\text{real}} = \frac{\delta_{\text{crisis}}}{\beta x_M^* + \delta_{\text{crisis}}}$ yields $u = \frac{\delta_{\text{crisis}}(1+h_0+\tilde{\xi}u^2)}{\beta M_0 + \delta_{\text{crisis}}(1+h_0+\tilde{\xi}u^2)}$. Rearranging algebraically isolates $L(u) = \frac{\beta M_0}{\delta_{\text{crisis}}}\frac{u}{1-u}$ on the left and $R(u) = 1 + h_0 + \tilde{\xi}u^2$ on the right. Since $L(0) = 0 < R(0) = 1 + h_0$ and $\lim_{u\to 1^-} L(u) = +\infty > R(1)$, at least one root exists in $(0,1)$. Because $L(u)$ is strictly increasing with $L''(u) = \frac{2\beta M_0}{\delta_{\text{crisis}}(1-u)^3}$ and $R(u)$ is quadratic, varying $\tilde{\xi}$ across the tangency condition produces a classic fold (saddle-node) bifurcation with two stable attractors ($u_1^* \ll 1$ and $u_3^* \to 1$) separated by an unstable tipping threshold u_2^*. □

Part III

Module 3: The Valorization Operator, the Conservation Contradiction, and the Mechanics of Surplus-Value (Volume I, Chapters 4–11)

10 Topological Inversion of the Circuit: $C-M-C$ vs. $M-C-M'$ (Chapter 4)

In Part II of *Capital* (Chapters 4–6), Marx introduces the **General Formula for Capital**:

$$M - C - M' \quad \text{where } M' = M + \Delta M,\ \Delta M > 0 \tag{26}$$

Whereas simple circulation $\hat{\mathcal{M}}_{i\to j} = \hat{\mathcal{B}}_j \circ \hat{\mathcal{S}}_i : \mathcal{U}_i \to \mathcal{U}_j$ maps between qualitatively distinct physical cones under quantitative value equivalence ($V_i = V_j$), the capital circuit $\hat{\mathcal{K}}_i = \hat{\mathcal{S}}'_i \circ \hat{\mathcal{P}} \circ \hat{\mathcal{B}}_i : \mathcal{U}_m \to \mathcal{U}_m$ is an endomorphism on the 1-dimensional Universal Equivalent space $\mathcal{U}_m$. Because its initial and terminal spaces are qualitatively identical ($\mathbb{R}_{\geq 0}\cdot\mathbf{e}_m$), the transformation is tautological unless it achieves a strictly positive scalar dilation $\Delta M > 0$ (**Surplus-Value** s).

11 The Impossibility Theorem of Pure Circulation (Chapter 5)

In Chapter 5 (*"Contradictions in the General Formula"*), Marx proves that surplus-value ($\Delta V > 0$) cannot arise inside the sphere of circulation, even under arbitrary non-equivalent exchange ("buying cheap and selling dear").

Theorem 11.1 (Conservation of Value in Pure Circulation under Skew-Symmetric Markups)**.** *Let $\mathcal{A} = \{1, 2, \ldots, N\}$ be a closed system of trading agents. Prior to trade, agent $\alpha \in \mathcal{A}$ holds commodity bundle $\mathbf{q}_\alpha \in \bigoplus_{i\in\mathcal{T}_0} \mathcal{U}_i$ and money stock $m_\alpha \mathbf{e}_m \in \mathcal{U}_m$, with initial total value:*

$$V_\alpha(0) = \sum_{i\in\mathcal{T}_0} \lambda_i q_{\alpha,i} + \lambda_m m_\alpha \tag{27}$$

Suppose any pair of agents (α, β) executes bilateral trades of commodity i in physical volume $X^{(i)}_{\alpha\beta} = -X^{(i)}_{\beta\alpha}$ ($X^{(i)}_{\alpha\beta} > 0$ denoting net sale from α to β) at an arbitrary negotiated market price $p^{(i)}_{\alpha\beta} = \frac{\lambda_i}{\lambda_m}(1 + \delta^{(i)}_{\alpha\beta})$, where $\delta^{(i)}_{\alpha\beta} \in (-1, \infty)$ is an arbitrary bilateral markup/markdown parameter. Then:

(1) ***Individual Value Shift:*** *Agent α's post-exchange value changes by:*

$$\Delta V_\alpha = V_\alpha(1) - V_\alpha(0) = \sum_{\beta\in\mathcal{A}} \sum_{i\in\mathcal{T}_0} \lambda_i \delta^{(i)}_{\alpha\beta} X^{(i)}_{\alpha\beta} \tag{28}$$

(2) ***Global Conservation Invariance:*** *Across the entire economy (or any closed trading sub-coalition), aggregate value is strictly invariant:*

$$\Delta V_{total} = \sum_{\alpha\in\mathcal{A}} \Delta V_\alpha \equiv 0 \tag{29}$$

(3) ***Zero-Sum Uniform Surcharge:*** *If every seller imposes a uniform proportional markup $\delta^{(i)}_{\alpha\beta} = \delta > 0$ when selling and pays δ when buying with balanced trade ($\sum_{i,\beta} \lambda_i X^{(i)}_{\alpha\beta} = 0$), then $\Delta V_\alpha = 0$ for every individual agent $\alpha \in \mathcal{A}$.*

Proof. When α sells $X^{(i)}_{\alpha\beta}$ units of commodity i to β at price $p^{(i)}_{\alpha\beta}$, α surrenders commodity value $\lambda_i X^{(i)}_{\alpha\beta}$ and receives money value $\lambda_m p^{(i)}_{\alpha\beta} X^{(i)}_{\alpha\beta} = \lambda_i(1+\delta^{(i)}_{\alpha\beta})X^{(i)}_{\alpha\beta}$, yielding a net bilateral value transfer $\pi^{(i)}_{\alpha\beta} = \lambda_i \delta^{(i)}_{\alpha\beta} X^{(i)}_{\alpha\beta}$. For counterparty β, the net value change is $\pi^{(i)}_{\beta\alpha} = -\lambda_i \delta^{(i)}_{\alpha\beta} X^{(i)}_{\alpha\beta} = -\pi^{(i)}_{\alpha\beta}$. Thus, for each commodity i, the value-transfer matrix $\Pi^{(i)} = [\pi^{(i)}_{\alpha\beta}]_{\alpha,\beta\in\mathcal{A}}$ is strictly skew-symmetric: $(\Pi^{(i)})^\top = -\Pi^{(i)}$. Summing over all agents yields:

$$\Delta V_{\text{total}} = \sum_{i\in\mathcal{T}_0} \mathbf{1}^\top \Pi^{(i)} \mathbf{1} = \frac{1}{2} \sum_{i\in\mathcal{T}_0} \mathbf{1}^\top \left(\Pi^{(i)} + \left(\Pi^{(i)}\right)^\top \right) \mathbf{1} = 0 \tag{30}$$

proving that pure circulation is a strictly conservative (measure-preserving) operator on total value. □

12 Labor-Power vs. Living Labor: Stock-Flow Operator Splitting (Chapter 6)

To satisfy the boundary-value problem of Chapter 5—where surplus-value must be generated while obeying strict equivalence ($\delta_{\alpha\beta} \equiv 0$) at every market node—the capitalist must find on the market a commodity whose **Use-Value is itself the active source of Value**. Marx resolves this via a rigorous dimensional distinction between **Labor-Power** (*Arbeitskraft*, a stock commodity) and **Living Labor** (*Arbeit*, a continuous flow operator).

Definition 12.1 (Labor-Power $lp \in \mathcal{T}_0$ and its Stock Reproduction Value λ_{lp})**.** *Let $lp \in \mathcal{T}_0$ denote the commodity **Labor-Power**, sold for a discrete contract period (1 working day) under the **Double Freedom Precondition** (the worker ω possesses legal self-ownership over $\mathcal{U}_{lp}$ but holds zero means of production, $\mathbf{q}^{means}_\omega = \mathbf{0}$). Let $\mathbf{b} = (b_1, b_2, \ldots, b_n)^\top \in \mathbb{R}^n_{\geq 0}$ be the culturally and historically determined daily subsistence bundle required to reproduce the worker's capacity, family, and skill. The unit stock value of Labor-Power is the inner product:*

$$\lambda_{lp} = V(\mathbf{b}) = \boldsymbol{\lambda}^\top \mathbf{b} = \sum_{i\in\mathcal{T}_0} \lambda_i b_i \quad \textit{[measured in hours/day]} \tag{31}$$

Definition 12.2 (Living Labor as a Value-Generating Flow Operator $\hat{\mathcal{L}}_T$)**.** *The consumption of the use-value of Labor-Power inside the production process $\mathcal{P}$ over a working day of duration $T \in (0, T_{\max}]$ is **Living Labor**—a continuous flow operator $\hat{\mathcal{L}}_T$ projecting concrete labor through the Real Abstraction Operator $\mathcal{A}$ at normalized socially average intensity $\iota(t) \equiv 1$:*

$$\Delta V(T) = \text{Use-Value}(lp)\big|_{[0,T]} = \int_0^T \iota(t)\, dt = T \quad \textit{[measured in hours/day]} \tag{32}$$

Theorem 12.1 (Resolution of the Conservation Contradiction via Stock-Flow Defect)**.** *Under strict equivalence of exchange in all commodity and labor markets ($\delta_{\alpha\beta} \equiv 0$), the net surplus-value s generated per worker per day is the **Stock-Flow Defect** between the flow action of $\hat{\mathcal{L}}_T$ and the stock reproduction value λ_{lp}:*

$$s(T, \mathbf{b}, \boldsymbol{\lambda}) = \Delta V(T) - \lambda_{lp} = \int_0^T \iota(t)\, dt - \boldsymbol{\lambda}^\top \mathbf{b} = T - \sum_{i\in\mathcal{T}_0} \lambda_i b_i \tag{33}$$

Whenever social productivity satisfies $\boldsymbol{\lambda}^\top \mathbf{b} < T$, strictly positive surplus-value $s > 0$ is generated without violating equivalent exchange at any boundary of circulation.

13 Bifurcation of the Production Process: Constant vs. Variable Capital (Chapters 7–9)

Consider the production of Q_j units of commodity j over 1 working day using physical means of production $\mathbf{a}_j = (a_{1j}, \dots, a_{nj})^\top \in \mathbb{R}^n_{\geq 0}$ and $L_j > 0$ living workers each working T hours/day.

Proposition 13.1 (Dual Operator Action of Concrete vs. Abstract Labor in Production)**.** *During the production interval $[0, T]$, living labor executes two simultaneous, orthogonal mathematical operations:*

(i) ***Value-Preservation Homomorphism (Constant Capital c_j):*** *Through its concrete useful character, labor preserves and transfers the pre-existing value of the consumed means of production $\mathbf{a}_j$ into the output $Q_j\mathbf{e}_j$ as an exact scalar invariant:*

$$c_j = V_{transferred}(\mathbf{a}_j) = \boldsymbol{\lambda}^\top \mathbf{a}_j = \sum_{i \in \mathcal{T}_0} \lambda_i a_{ij} \implies \Delta c_j \equiv 0 \tag{34}$$

(ii) ***Value-Dilation Operator (Variable Capital v_j):*** *Through its abstract social character, living labor injects fresh value $L_j T$, replacing its advanced variable capital $v_j = L_j \lambda_{lp}$ and generating surplus-value s_j:*

$$v_j \xrightarrow{\hat{L}_T} v_j + s_j = L_j \lambda_{lp} + L_j \left(T - \lambda_{lp}\right) = L_j T \implies \Delta v_j = s_j = L_j \left(T - \boldsymbol{\lambda}^\top \mathbf{b}\right) > 0 \tag{35}$$

The total value of output $C'_j = Q_j \mathbf{e}_j$ is the affine functional in T:

$$W_j(T) = c_j + v_j + s_j(T) = \underbrace{\boldsymbol{\lambda}^\top \mathbf{a}_j}_{\text{Dead Labor } (c_j)} + \underbrace{L_j \left(\boldsymbol{\lambda}^\top \mathbf{b}\right)}_{\text{Necessary Labor } (v_j)} + \underbrace{L_j \left(T - \boldsymbol{\lambda}^\top \mathbf{b}\right)}_{\text{Surplus Labor } (s_j)} \tag{36}$$

Definition 13.1 (Rate of Surplus-Value / Rate of Exploitation e)**.** *The* ***Rate of Surplus-Value*** *e is the fractional dilation of variable capital alone:*

$$e \coloneqq \frac{\Delta v}{v} = \frac{s}{v} = \frac{T - \lambda_{lp}}{\lambda_{lp}} = \frac{T - \boldsymbol{\lambda}^\top \mathbf{b}}{\boldsymbol{\lambda}^\top \mathbf{b}} = \frac{t_s}{t_n} \tag{37}$$

partitioning the working day $T = t_n + t_s$ into Necessary Labor Time $t_n = \frac{T}{1+e}$ and Surplus Labor Time $t_s = \frac{e}{1+e} T$.

14 Differential Geometry of Exploitation, the Working Day, and the Three Laws of Surplus-Value Mass (Chapters 10–11)

14.1 Total Differential Decomposition of Exploitation Strategies

Taking the total differential of $e(T, \boldsymbol{\lambda}, \mathbf{b}) = \frac{T}{\boldsymbol{\lambda}^\top \mathbf{b}} - 1$ on the production manifold yields:

$$de = \underbrace{\left(\frac{1}{\lambda_{lp}}\right) dT}_{\substack{\text{Absolute Surplus-Value} \\ \text{(Ch. 10: Working Day Elongation)}}} + \underbrace{\left(-\frac{T}{\lambda_{lp}^2} \mathbf{b}^\top d\boldsymbol{\lambda}\right)}_{\substack{\text{Relative Surplus-Value} \\ \text{(Ch. 12: Productivity Revolution)}}} + \underbrace{\left(-\frac{T}{\lambda_{lp}^2} \boldsymbol{\lambda}^\top d\mathbf{b}\right)}_{\substack{\text{Real Wage Compression} \\ \text{(Depression below } \lambda_{lp})}} \tag{38}$$

14.2 The Antinomy of the Working Day and Biological Spoliation (Chapter 10)

In Chapter 10, Marx shows that when the working day T exceeds a physiological regeneration threshold T_{phys}, overwork induces premature exhaustion and shortens the worker's productive lifespan. We formalize this via the **True Biological Reproduction Cost Functional**:

$$\Lambda_{lp}^{\text{true}}(T) = \lambda_{lp} + \gamma \max(0,\, T - T_{\text{phys}})^2\,, \quad \gamma > 0 \tag{39}$$

Whenever $T > T_{\text{phys}}$ while the daily wage remains fixed at λ_{lp}, capital covertly liquidates the worker's biological stock ($\Lambda_{lp}^{\text{true}}(T) - \lambda_{lp} > 0$), violating long-run equivalent exchange. Because "between equal rights, force decides," the working day $T(t) \in (\lambda_{lp}, T_{\max})$ evolves via the **Class-Struggle Differential Game**:

$$\dot{T} = \kappa_K \frac{\partial s}{\partial T}(T_{\max} - T) - \kappa_W \frac{\partial \Lambda_{lp}^{\text{true}}}{\partial T}(T - \lambda_{lp}) = \kappa_K (T_{\max} - T) - 2\kappa_W \gamma \max(0,\, T - T_{\text{phys}})\,(T - \lambda_{lp}) \tag{40}$$

where $\kappa_K > 0$ and $\kappa_W > 0$ denote the bargaining/coercive force parameters of Capital and Labor/State factory legislation, respectively.

14.3 Marx's Three Laws of the Mass of Surplus-Value (Chapter 11)

Theorem 14.1 (The Three Laws of Chapter 11 and the Absolute Substitution Barrier)**.** *Let an enterprise employ L workers with variable capital per worker λ_{lp} and working day $T \le T_{\max} < 24$ hrs:*

(1) ***First Law (Proportionality to Variable Capital & Rate of Exploitation):*** *The aggregate Mass of Surplus-Value S is given by:*

$$S(L, T, \lambda_{lp}) = V_{total} \cdot e = L \cdot \lambda_{lp} \left(\frac{T - \lambda_{lp}}{\lambda_{lp}} \right) = L\,(T - \lambda_{lp}) \tag{41}$$

(2) ***Second Law (Iso-Surplus Compensation Differential):*** *Along an iso-surplus manifold $S(L,T) = S_0$, a reduction in the living workforce $dL < 0$ (via mechanization) requires a compensating relative increase in the working day (or rate of exploitation):*

$$dS = (T - \lambda_{lp})dL + L\,dT = 0 \implies \frac{dT}{T - \lambda_{lp}} = -\frac{dL}{L} \tag{42}$$

(3) ***Third Law (Absolute Physical Barrier to Labor-Saving Substitution):*** *Because T is bounded above by $T_{\max} < 24$ hrs, there exists a strictly positive critical workforce threshold:*

$$L_{crit} \coloneqq \frac{S_0}{T_{\max} - \lambda_{lp}} > 0 \tag{43}$$

such that for any $L < L_{crit}$, $S(L,T) < S_0$ for all admissible $T \in (\lambda_{lp}, T_{\max}]$. That is, beyond L_{crit}, no extension of the working day or intensification of exploitation can compensate for the expulsion of living labor from production.

Proof. Parts (1) and (2) follow immediately from definition and total differentiation of $S(L,T) = L(T - \lambda_{lp})$. For Part (3), since $T \le T_{\max}$, we have $S(L,T) = L(T - \lambda_{lp}) \le L(T_{\max} - \lambda_{lp})$. If $L < L_{\text{crit}} = \frac{S_0}{T_{\max} - \lambda_{lp}}$, then $S(L,T) \le L(T_{\max} - \lambda_{lp}) < S_0$, establishing the absolute upper bound that underpins the Tendency of the Rate of Profit to Fall in Volume III. □

Part IV
Module 4: Relative Surplus-Value, Superprofit Dynamics, Co-operation, and Machinery (Volume I, Chapters 12–15)

15 The Micro-Macro Paradox of Relative Surplus-Value (Chapter 12)

In Part III of *Capital*, the rate of surplus-value $e = \frac{T-\lambda_{lp}}{\lambda_{lp}}$ was increased extensively by lengthening the working day T ($dT > 0$, Absolute Surplus-Value). In Part IV (Chapters 12–15), T is held fixed ($dT = 0$) and e is increased intensively by contracting Necessary Labor Time $t_n = \lambda_{lp}$ via revolutions in the productivity of labor (**Relative Surplus-Value**).

Marx identifies a foundational micro-macro paradox in Chapter 12: no individual capitalist introduces productivity-enhancing techniques in order to cheapen the worker's subsistence bundle $\mathbf{b}$ and lower λ_{lp}. Rather, the individual firm innovates to capture transient **Extra Surplus-Value (*Extra-Mehrwert* / Superprofit)** by driving its *individual value* below the *social value* λ_j.

15.1 Individual Value vs. Social Value and *Extra-Mehrwert*

Recall from Module 1 that the Social Unit Value (SNLT) of commodity $j \in \mathcal{T}_0$ is the expectation of individual unit labor times over the sectoral producer measure $\mu_j(\omega, t)$:

$$\lambda_j(t) = \mathcal{A}(\mathcal{L}_j) = \int_{\mathcal{L}_j} \tau_j(\omega, t)\, d\mu_j(\omega, t) = \bar{c}_j(t) + \bar{l}_j(t) \tag{44}$$

where $\bar{c}_j$ and $\bar{l}_j$ denote the socially average transferred constant capital and direct living labor per physical unit $\mathbf{e}_j$.

Definition 15.1 (Individual Value and Productivity Multiplier)**.** *Let an innovating enterprise $\hat{\omega} \in \mathcal{L}_j$ introduce an advanced technique that reduces its direct living labor requirement per physical unit to $l_{\hat{\omega}} < \bar{l}_j$, defining a* ***productivity potentiation multiplier****:*

$$\theta_{\hat{\omega}}(t) \coloneqq \frac{\bar{l}_j(t)}{l_{\hat{\omega}}} > 1 \tag{45}$$

The enterprise's ***individual unit value*** $\lambda_{j,ind}(\hat{\omega}) = c_{\hat{\omega}} + l_{\hat{\omega}}$ *satisfies* $\lambda_{j,ind}(\hat{\omega}) < \lambda_j(t)$.

Theorem 15.1 (Generation and Competitive Dissipation of Extra Surplus-Value)**.** *Assume commodities of type j sell at the uniform social market price $p_j(t) = \lambda_j(t)/\lambda_m$ (with $c_{\hat{\omega}} = \bar{c}_j$). Then:*

(1) ***Potentiated Labor Action:*** *One hour of concrete labor expended inside enterprise $\hat{\omega}$ projects onto the social value manifold as* ***potentiated labor*** *(potenzierte Arbeit) of magnitude $\theta_{\hat{\omega}} > 1$. Over a working day of length T, each worker in $\hat{\omega}$ produces $\theta_{\hat{\omega}}(T/\bar{l}_j)$ units and realizes net social value added $\Delta V_{\hat{\omega}}(T) = \theta_{\hat{\omega}} T > T$.*

(2) ***Extra Surplus-Value Decomposition:*** *The total surplus-value realized per worker in firm $\hat{\omega}$ decomposes into normal sectoral surplus-value $s_0 = T - \lambda_{lp}$ plus a strictly positive* ***Extra Surplus-Value*** *$\Delta s_{\hat{\omega}}$:*

$$s_{\hat{\omega}}(t) = \theta_{\hat{\omega}}(t)T - \lambda_{lp} = \underbrace{(T - \lambda_{lp})}_{\text{Normal } s_0} + \underbrace{(\theta_{\hat{\omega}}(t) - 1)\,T}_{\text{Extra Surplus-Value } \Delta s_{\hat{\omega}} > 0} \tag{46}$$

(3) ***Measure Transport and Asymptotic Dissipation:*** *Under competitive imitation, rival enterprises adopt the superior technique, governing the sectoral measure $\mu_j(\omega, t)$ via the advection-diffusion (Fokker-Planck) transport equation $\partial_t \mu_j + \nabla_\tau \cdot (v_{adopt} \mu_j) = \frac{\sigma^2}{2} \Delta_\tau \mu_j$. As $\mu_j(\omega, t)$ concentrates on $l_{\hat{\omega}}$:*

$$\lim_{t\to\infty} \bar{l}_j(t) = l_{\hat{\omega}} \implies \lim_{t\to\infty} \theta_{\hat{\omega}}(t) = 1 \implies \lim_{t\to\infty} \Delta s_{\hat{\omega}}(t) = 0 \tag{47}$$

Thus, Extra Surplus-Value is strictly self-extinguishing under competitive diffusion.

Proof. In T hours, a worker in $\hat{\omega}$ produces $q_{\hat{\omega}} = T/l_{\hat{\omega}}$ physical units. Selling at the social unit value $\lambda_j = \bar{c}_j + \bar{l}_j$ yields gross revenue $q_{\hat{\omega}}(\bar{c}_j + \bar{l}_j)$. Subtracting transferred constant capital $q_{\hat{\omega}} c_{\hat{\omega}} = q_{\hat{\omega}} \bar{c}_j$ leaves net value added $q_{\hat{\omega}} \bar{l}_j = T(\bar{l}_j / l_{\hat{\omega}}) = \theta_{\hat{\omega}} T$. Deducting the wage λ_{lp} yields $s_{\hat{\omega}} = \theta_{\hat{\omega}} T - \lambda_{lp} = (T - \lambda_{lp}) + (\theta_{\hat{\omega}} - 1)T$. As the measure $\mu_j(\omega, t) \rightharpoonup \delta(\tau - \lambda_{j,\mathrm{ind}}(\hat{\omega}))$ weakly as $t \to \infty$, $\bar{l}_j(t) \to l_{\hat{\omega}}$, driving $\theta_{\hat{\omega}}(t) \to 1$ and $\Delta s_{\hat{\omega}}(t) \to 0$. □

15.2 Departmental Spectral Partition: Wage-Goods vs. Luxury Goods

Proposition 15.1 (Departmental Sensitivity of the General Rate of Exploitation)**.** *Let $A = [a_{ik}]_{i,k\in\mathcal{T}_0}$ be the technical input matrix and $\mathbf{l} = (l_1, \ldots, l_n)^\top$ the direct living labor vector, so that $\boldsymbol{\lambda}^\top = \mathbf{l}^\top (I - A)^{-1}$ and $\lambda_{lp} = \mathbf{l}^\top (I - A)^{-1} \mathbf{b}$. Partition $\mathcal{T}_0 = \mathcal{T}_{wage} \sqcup \mathcal{T}_{lux}$, where $\mathcal{T}_{wage} = \{k \in \mathcal{T}_0 \mid [(I - A)^{-1}\mathbf{b}]_k > 0\}$ (direct and indirect wage-goods) and $\mathcal{T}_{lux} = \{k \in \mathcal{T}_0 \mid [(I - A)^{-1}\mathbf{b}]_k = 0\}$ (pure luxury goods). Then:*

$$\forall k \in \mathcal{T}_{wage}: \quad \frac{\partial e}{\partial l_k} = -\frac{T}{\lambda_{lp}^2} \left[(I - A)^{-1}\mathbf{b}\right]_k < 0, \qquad \forall k \in \mathcal{T}_{lux}: \quad \frac{\partial e}{\partial l_k} \equiv 0 \tag{48}$$

Hence, technological revolutions generate systemic Relative Surplus-Value ($de > 0$) if and only if they occur within the transitive input closure of the worker's subsistence bundle $\mathcal{T}_{wage}$.

16 Superadditivity of Co-operation and the Division of Labor (Chapters 13 & 14)

16.1 Co-operation as a Central Limit Equalizer and Superadditive Operator (Chapter 13)

Proposition 16.1 (Statistical Equalization and Collective Productive Power in Co-operation)**.** *When a capitalist simultaneously employs L workers in a coordinated workshop:*

(i) ***Law of Large Numbers Realization of SNLT:*** *If individual worker productivity $\tau(\omega_r)$ has mean λ_j and finite variance $\sigma_\tau^2 > 0$, the firm's sample unit labor time $\bar{\tau}_L = \frac{1}{L}\sum_{r=1}^{L} \tau(\omega_r)$ satisfies $\mathrm{Var}(\bar{\tau}_L) = \sigma_\tau^2 / L \to 0$ as $L \to \infty$. Thus, a sufficiently large workforce L converts the stochastic individual labor process into the deterministic social law of value inside the factory.*

(ii) ***Superadditivity of the Collective Worker (Gesamtarbeiter):*** *Whereas isolated labor is strictly additive ($\hat{F}_{iso}(L) = L\bar{f}$), coordinated co-operation induces a superadditive physical output functional $\hat{F}_{coop}(L) = L^{1+\eta}\bar{f}$ ($\eta > 0$) and subadditive fixed constant capital overhead $C_{fixed}(L) = c_0 L^\nu$ ($0 < \nu < 1$). Because capital pays $L\lambda_{lp}$ for L isolated labor-powers, the superadditive output wedge $(L^{1+\eta} - L)\bar{f}$ is appropriated gratis as surplus-value.*

16.2 Manufacture as a Directed Acyclic Task Graph and the Babbage Principle (Chapter 14)

In Manufacture (Chapter 14), a craft process is decomposed into K specialized partial operations $\mathcal{O} = \{o_1, \ldots, o_K\}$ forming a **Directed Acyclic Graph (DAG)** $\mathcal{G}_{\mathrm{man}} = (\mathcal{O}, E_{\mathrm{flow}})$.

Proposition 16.2 (Flow Conservation and Skill Devaluation in Manufacture). *Let $\tau_a > 0$ be the cycle time of partial operation $o_a \in \mathcal{O}$ and $\epsilon_1 > \epsilon_2 > \cdots > \epsilon_K = 0$ be the amortized skill-training reproduction costs of the respective operations:*

(1) ***Marx's Law of Proportional Subgroup Sizes:*** *Eliminating intermediate inventory bottlenecks and inter-task switching dead-time (Poren der Arbeitszeit) requires subgroup headcounts $(L_1, \ldots, L_K)$ to satisfy the exact integer flow invariant:*

$$\frac{L_1}{\tau_1} = \frac{L_2}{\tau_2} = \cdots = \frac{L_K}{\tau_K} = \Phi_{line} \implies L_a^* = m \cdot \frac{\tau_a}{\gcd(\tau_1, \ldots, \tau_K)} \quad (m \in \mathbb{N}) \tag{49}$$

(2) ***Structural Devaluation of Labor-Power via Skill Hierarchy:*** *By restricting each worker subgroup L_a to a single micro-operation o_a, capital pays only the task-specific skill cost ϵ_a rather than the full artisan training cost ϵ_1, strictly lowering the average value of labor-power even at constant wage-good values $\boldsymbol{\lambda}$:*

$$\bar{\lambda}_{lp}^{man} = \lambda_{lp}^{(0)} + \frac{\sum_{a=1}^{K} L_a \epsilon_a}{\sum_{a=1}^{K} L_a} < \lambda_{lp}^{(0)} + \epsilon_1 = \bar{\lambda}_{lp}^{craft} \tag{50}$$

17 Machinery, Moral Depreciation, and the Bridge to Stochastic Mean Field Accumulation (Chapter 15)

17.1 The Capitalist Boundary Condition for Machine Adoption (Chapter 15, Section 2)

Theorem 17.1 (Divergence between Social and Capitalist Automation Boundaries). *Let a machine require C_{mach} hours of SNLT to produce and suppose it displaces L_{disp} workers over its lifespan, where each worker works $T = \lambda_{lp} + s$ hours.*

(i) ***Social Labor-Saving Criterion:*** *The machine reduces total human labor expended on society's output if and only if:*

$$C_{mach} < L_{disp} \cdot T = L_{disp}\,(\lambda_{lp} + s) \tag{51}$$

(ii) ***Capitalist Cost-Reduction Criterion:*** *Because the capitalist pays only for variable capital λ_{lp} and not for living labor T, the machine is adopted under capitalism if and only if:*

$$C_{mach} < L_{disp} \cdot \lambda_{lp} = \frac{L_{disp} \cdot T}{1 + e} \tag{52}$$

(iii) ***The Sweatshop Stagnation Wedge:*** *For any machine satisfying $L_{disp}\lambda_{lp} \leq C_{mach} < L_{disp}T$ (a wedge of width $L_{disp} \cdot e\lambda_{lp}$), automation is socially labor-saving but is rejected by capital. Thus, depressed wages ($\lambda_{lp} \downarrow$) and high exploitation ($e \uparrow$) structurally block technological modernization.*

17.2 Moral Depreciation and the Compulsion to Extend and Intensify Labor (Chapter 15, Section 3)

Theorem 17.2 (Moral Depreciation Hazard and the Elongation/Condensation of the Working Day). *Let a machine of initial value C_0 transfer value into output at rate βT per day while facing continuous technological obsolescence (**Moral Depreciation**, moralischer Verschleiß) at hazard rate $\delta_{moral} > 0$, which erodes its replacement value as $C(t) = C_0 e^{-\delta_{moral} t}$. The cumulative fraction of C_0 successfully transferred before physical exhaustion at $\tau_{phys}(T) = C_0/(\beta T)$ is:*

$$\mathcal{R}_{transfer}(T) = \int_0^{\frac{C_0}{\beta T}} \frac{\beta T}{C_0} e^{-\delta_{moral} t}\, dt = \frac{\beta T}{\delta_{moral} C_0} \left(1 - \exp\left(-\frac{\delta_{moral} C_0}{\beta T}\right)\right) \tag{53}$$

Because $\frac{\partial \mathcal{R}_{transfer}}{\partial T} > 0$ *for all* $\delta_{moral} > 0$*, moral depreciation creates an endogenous compulsion to maximize daily machine utilization* $T \to 24$ *hrs. When state legislation caps* $T \leq T_{legal}$*, capital substitutes* ***intensive magnitude (condensation of labor*** $\iota > 1$***)*** *for extensive duration* T*, injecting effective value* $\Delta V = \iota T_{legal}$ *per day.*

17.3 Synthesis: Microeconomic Foundations of the Stochastic Mean Field Game (MFG)

Combining Theorem 15.1 (*Extra-Mehrwert*) and Theorem 17.2 (*Moral Depreciation*) yields the exact micro-foundation for the Volume III **Tendency of the Rate of Profit to Fall (TRPF)** modeled as a coupled Hamilton-Jacobi-Bellman (HJB) and Fokker-Planck Mean Field Game:

1. Each enterprise $i \in [0, 1]$ controls its automation rate $u_i(t)$ governing its individual organic composition $\omega_i(t) = C_{i,t}/V_{i,t}$ subject to moral depreciation δ_{moral} and adoption noise σ_1:

$$d\omega_i(t) = (u_i(t) - \delta_{\text{moral}}\omega_i(t))\, dt + \sigma_1 \omega_i(t)\, dW_{i,t} \tag{54}$$

2. Enterprise i's instantaneous payoff combines the baseline valorization rate $\frac{e}{\omega_i+1}$ with the Chapter 12 *Extra-Mehrwert* superprofit term proportional to its technological lead over the Mean Field average $\bar{\omega}_t = \int_{\mathbb{R}_+} \omega\, m(\omega, t)\, d\omega$:

$$r_i(\omega_i, \bar{\omega}_t) = \frac{e(\bar{\omega}_t)}{\omega_i + 1} + \chi\left(\frac{\omega_i - \bar{\omega}_t}{\bar{\omega}_t + 1}\right) - \frac{\phi}{2}u_i^2 \tag{55}$$

3. While each atomic firm chooses $u_i^*(t) > 0$ to capture $\chi(\omega_i - \bar{\omega}_t) > 0$, the forward Kolmogorov (Fokker-Planck) transport of the capital distribution $m(\omega, t)$ drives $\bar{\omega}_t$ upward. In the symmetric Mean Field Equilibrium ($\omega_i = \bar{\omega}_t$), the superprofit term vanishes identically ($\chi(\bar{\omega}_t - \bar{\omega}_t) \equiv 0$), trapping the aggregate economy in a higher organic composition $\bar{\omega}_t$ and a secularly declining general rate of profit $r(\bar{\omega}_t) = \frac{e(\bar{\omega}_t)}{\bar{\omega}_t+1}$.